

Feature Status Ledger

Revision: 1 **Last modified:** 2026-06-15T15:27:34Z **Authority:** constitution Â§11.4.153 (per-feature Status + video-recording confirmation), composes Â§11.4.44 (revision header), Â§11.4.45 (integration-status-doc + closed status vocabulary), Â§11.4.56 (Status_Summary two-audience companion), Â§11.4.65 (HTML+PDF export) + Â§11.4.153 DOCX, Â§11.4.107 (no frozen-frame video proof), Â§11.4.6 (no-guessing â€” every row reconciled against real code). **Scope:** Full feature surface of the yt-dlp Podman/Docker orchestration around yt-dlp â€” every service, client surface, CLI script, landing route, dashboard component, test/challenge suite, and every PLANNED-but-unbuilt feature. **Maintainer:** project conductor.

How to read this ledger

- **Implementation** â€” Implemented (code present + wired), Partial (code present, incomplete/edge-gaps), Planned (not yet built).
- **Validation/Verification** â€” closed status vocabulary per Â§11.4.45: PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED. Because no clean-baseline full-suite run with captured evidence was executed in the session that produced this ledger, verdicts that would require live captured runtime evidence are recorded PENDING_FORENSICS (per Â§11.4.6 â€” an unproven PASS is forbidden; the code is present and test files exist, but a green-with-captured-evidence run is not yet on record here). Planned rows carry SKIP (no feature to validate yet).
- **Video-Confirmation** â€” per Â§11.4.153 every user-visible confirmed claim **MUST** be backed by a recorded real-use video. **NO feature has a video recording yet.** Therefore **EVERY** rowâ€™s Video-Confirmation cell is PENDING â€” not yet recorded, without exception. None is â€œconfirmedâ€.

HONEST FACT (Â§11.4.6 / Â§11.4.153): Not a single feature in this project has a real-use video recording at the time of this revision. Every Video-Confirmation cell below reads PENDING â€” not yet recorded. No row claims video confirmation.

OPERATOR-BLOCKED rows (top per Â§11.4.45)

No feature in this project is currently OPERATOR-BLOCKED. (The VPN profile requires operator-supplied .ovpn credentials and vpn-auth.txt, but the VPN *code path* is implemented and testable in compose-config form without live credentials, so it is classified PENDING_FORENSICS, not OPERATOR-BLOCKED.)

1. SERVICES (docker-compose.yml)

Feature	Component	Category	Implementation	Wiring
VPN tunnel (dperson/openvpn-client, NET_ADMIN, /dev/	openvpn-yt-dlp	Networking / Service	Implemented (vpn profile)	Reachable: metube + yt-dlp-cli-netns via network_mode: service yt-dlp

Feature	Component Category			Implementation Wiring	
net/tun, port 3130, healthcheck ping 8.8.8.8) Download queue/history API + minimal UI (ghcr.io/alexta69/metube:latest, port 8088 direct / netns under VPN)	metube / metube-direct	Backend API / Service	Implemented	Reachable: dashboard nginx + landing :8081; \${DOWNLOAD_DIR} + ./me config volumes	
Flask cookie-auth onboarding + aborted-history ledger + proxy endpoints (landing/, ports 8087 vpn / 8086 no-vpn)	landing-vpn / landing-no-vpn	Web UI + API / Service	Implemented	Reachable: published port; proxies cookie-auth onboarding download to MeTube; writes /config aborted.json	
Sleeping CLI container for ./download (ghcr.io/jim60105/yt-dlp:pot, while true; do sleep 3600; done, cookie copy at startup)	yt-dlp-cli / yt-dlp-cli-vpn	CLI backend / Service	Implemented	Reachable: ./download execs yt-dlp/config, cookies, archive/downloads volumes	
Angular 17 + nginx dashboard (dashboard/, port 9090, /api/* proxied to MeTube + landing)	dashboard	Web UI / Service	Implemented (no-vpn profile only)	Reachable: nginx resolver + variable proxy_pass from entripoint.s	
Image auto-update (containrrr/watchtower:latest, schedule 0 0 */3 * * *, cleanup)	watchtower	Infra / Service	Implemented (docker profile)	Reachable: docker.sock mount; gated by com.centurylinklabs.watchtower labels	

2. CLIENT SURFACES

Feature	Component Category			Implementation Wiring	Tests Coverage
Dashboard web UI (download-form / queue / history / cookies / navbar, error-boundary, not-found)	dashboard	Web client	Implemented	Routed via app.routes.ts; calls /api/* through nginx	tests/test-dashboa e2e/tests/dashboard component *.spec.ts
Landing Flask UI (3-step cookie onboarding, drag-	landing	Web client	Implemented	INDEX_TEMPLATE in landing/app.py; auto-	tests/e2e/tests/ landing.spec.ts, te dashboard.sh

Feature	Component	Category	Implementation	Wiring	Tests Coverage
drop upload, freshness banner, services grid)				redirect to dashboard	
MeTube minimal UI (vendor original interface, dark theme)	metube	Web client	Implemented (vendor)	Served on :8088; linked as "Classic UI" from landing	tests/test-media-s
./download CLI (host script execs yt-dlp in sleeping container)	download	CLI client	Implemented	./download â†’ exec into yt-dlp-cli; runtime auto-detect	tests/test-media-s challenges/scripts download_completes

3. ROOT CLI SCRIPTS

Feature	Component	Category	Implementation	Wiring	Tests Coverage
Init environment (create dirs, validate .env, generate yt-dlp.conf, write vpn-auth.txt chmod 600)	./init	CLI script	Implemented	Runtime auto-detect via lib/container-runtime.sh	tests/test-unit.sh, tests/test-integration.sh, tests/test-s
Start services (reads USE_VPN, pulls images, brings up profile)	./start	CLI script	Implemented	compose up via detected runtime	tests/test-scenarios.sh, test-integration.sh
Start no-vpn (force no-vpn profile)	./start_no_vpn	CLI script	Implemented	compose --profile no-vpn	tests/test-scenarios.sh, challenges/scripts/no_vpn_profile_direct_acces
Stop services (all profiles, clean pods/networks)	./stop	CLI script	Implemented	compose down + cleanup	tests/test-scenarios.sh
Status summary (service + HTTP health)	./status	CLI script	Implemented	queries running containers + HTTP endpoints	tests/test-integration-real
Download helper (yt-	./download		Implemented		

Feature	Component	Category	Implementation	Wiring	Tests Coverage
dlp via exec)		CLI script		exec into yt-dlp-cli	tests/test-media-services.scripts/download_completes_
Check VPN (query ipinfo.io inside VPN container)	./check-vpn	CLI script	Implemented	exec into openvpn-yt-dlp	tests/test-vpn-smoke.sh
Restart services	./restart	CLI script	Implemented	stop + start sequence	tests/test-scenarios.sh
Cleanup containers/pods/networks	./cleanup	CLI script	Implemented	compose down + prune	tests/test-scenarios.sh,cha scripts/ container_restart_resilienc
Release preparation (commit + push)	./prepare-release.sh	CLI script	Implemented	git operations	bash -n syntax gate (make ci)
Auto-update setup (cron for Podman; Watchtower alt for Docker)	./setup-auto-update	CLI script	Implemented	writes cron entry / documents Watchtower	bash -n syntax gate
Update images (pull latest before start)	./update-images	CLI script	Implemented	compose pull via detected runtime	bash -n syntax gate

4. LANDING ROUTES (landing/app.py)

Feature	Component	Category	Implementation	Wiring
/ " onboarding page (3-step cookie flow, session id, dynamic service URLs)	landing	HTTP route	Implemented	render_template_string(INDEX_TEMPLA
/app " proxy GET to MeTube backend	landing	HTTP proxy route	Implemented	requests.get(METUBE_URL) streamed back
/api/ upload-	landing	HTTP API route	Implemented	_validate_cookie_file + POST to \${METUBE_URL}/upload-cookies

Feature	Component	Category	Implementation	Wiring
cookies (POST) â€” validate Netscape file (UTF-8 + recognised-domain gate) â†’ forward to MeTube				
/api/delete-cookies (POST) â€” remove /config/cookies.txt	landing	HTTP API route	Implemented	os.remove
/api/cookie-status (GET) â€” has_cookies + age + per-platform breakdown + MeTube reachability	landing	HTTP API route	Implemented	_summarize_cookies_by_platform + MeT fallback
/api/aborted-history (GET) â€” return persisted aborted list	landing	HTTP API route	Implemented	_read_aborted_history (lock-free read)
/api/aborted-history (POST) â€” append entry, flock-guarded, 60s idempotency	landing	HTTP API route	Implemented	_aborted_history_lock + _write_aborted_history
/api/aborted-history (DELETE) â€” remove specific urls or *	landing	HTTP API route	Implemented	locked read-modify-write
/api/profile-status (GET) â€” active compose profile	landing	HTTP API route	Implemented	reads ACTIVE_PROFILE env

Feature	Component	Category	Implementation	Wiring
+ vpn_active + MeTube reachability /health (GET) â€” service + MeTube reachability + timestamp	landing	HTTP API route	Implemented	requests.get(\${METUBE_URL}/history)
/logo.png (GET) â€” serve branding asset	landing	HTTP static route	Implemented	send_from_directory
/favicon.ico (GET) â€” 204 stub	landing	HTTP static route	Implemented	returns "", 204
/api/ delete- download (POST) â€” remove from history + optional file delete (path- traversal guarded)	landing	HTTP API route	Implemented	POST \${METUBE_URL}/delete + os.walk/ os.remove within DOWNLOAD_DIR

5. DASHBOARD COMPONENTS (dashboard/src/app)

Feature	Component	Category	Implementation	Wiring
Download form (url, quality select, format select, folder, add via /api/add, inline tracker + error state, re-enable after failure)	download-form	Angular component	Implemented	MeTubeService.addDown
Queue (STATE_META: pending / preparing / downloading / postprocessing / finished / error / aborted; cancel-all; record-and-close aborted)	queue	Angular component	Implemented	getHistoryPolling + recordAbortedItem
History (+ aborted-history merge via abortedToDownloadInfo, history retry, delete-with-file, batch + delete-all, refresh)		Angular component	Implemented	getHistory + getAbortedHistory + deleteDownloadWithFi
	cookies		Implemented	

Feature	Component	Category	Implementation	Wiring
Cookies (status badge fresh/ expiring/expired, age, MeTube online/offline, per-platform breakdown, upload, delete)		Angular component		<code>getCookieStatus + uploadCookies + deleteCookies</code>
Navbar (routerLinks Download/Queue/History/ Cookies; VPN / No-VPN / unknown profile badge; MeTube-unreachable badge)	navbar	Angular component	Implemented	<code>getProfileStatus</code>
Error boundary (renders loading / error+retry / empty / content states)	error-boundary	Angular component	Implemented	wrapped around list compone
Not-found (404 catch-all route)	not-found	Angular component	Implemented	<code>** route in app.routes.t</code>
Error interceptor (RxJS error surfacing, clears loading)	error-interceptor.service	Angular service	Implemented	<code>app.config.ts</code> provider
MeTube service (typed API: add/start/history/delete/clear/retry/cookies/profile/aborted/version/poll; no any)	metube.service	Angular service	Implemented	<code>HttpClient</code> to <code>/api/*</code>

6. TESTS / CHALLENGES (real test files reconciled)

Feature	Component	Category	Implementation	Wiring
Unit tests (pure-shell logic)	tests/test-unit.sh	Test suite	Implemented	<code>run-tests.sh -p unit</code>
Integration tests (containers up)	tests/test-integration.sh	Test suite	Implemented	<code>run-tests.sh -p integration</code>
Real-HTTP integration (no mocks, hits :9090/:8086/:8088)	tests/test-integration-realhttp.sh	Test suite	Implemented	services must be up
Scenario tests (lifecycle phases)	tests/test-scenarios.sh	Test suite	Implemented	<code>run-tests.sh -p scenario</code>
Error-phase tests	tests/test-errors.sh	Test suite	Implemented	<code>run-tests.sh -p erro</code>
Dashboard + landing integration	tests/test-dashboard.sh	Test suite	Implemented	services up
Dashboard operations (post-fix full automation)	tests/test-dashboard-operations.sh	Test suite	Implemented	services up

Feature	Component	Category	Implementation Wiring	
Aborted-history (incl.Â concurrent-posts no-500)	tests/test-aborted-history.sh	Test suite	Implemented	landing up
Cookie validator (<code>_validate_cookie_file</code>)	tests/test-cookie-validator.sh	Test suite	Implemented	landing app.py
Media-services compatibility (anti-bluff documented-failure)	tests/test-media-services.sh	Test suite	Implemented	yt-dlp-cli / metube up
Bulk operations	tests/test-bulk-operations.sh	Test suite	Implemented	services up
Add-all-platforms / add-download	tests/test-add-all-platforms.sh, tests/test-add-download.sh	Test suite	Implemented	metube up
VPN compose + smoke	tests/test-vpn-compose.sh, tests/test-vpn-smoke.sh	Test suite	Implemented	vpn profile
Chaos / resilience	tests/test-chaos.sh	Test suite	Implemented	services up
Constitution inheritance + meta-test	tests/test-constitution-inheritance.sh, tests/meta-test-constitution-inheritance.sh	Test suite	Implemented	repo tree
E2E (Playwright): dashboard, landing, cross-service, clear-history, delete-retry-cancel, cleanup-and-start	tests/e2e/	E2E suite	Implemented	run-e2e.sh + playwright.config.ts
Benchmark suite	tests/benchmark/run-benchmarks.sh	Benchmark suite	Implemented	services up
Cookie auth + youtube-download + validate-fix	tests/cookies/	Test suite	Implemented	metube + cookies
MeTube anti-bluff challenge bank (download-completes, queue-realtime, form-reenables)	tests/challenges/	Challenge bank	Implemented	run-metube-challenges.sh + metube-challenges.json
Root HelixQA challenge scripts (api-contract, container-restart-resilience, download-completes,	challenges/scripts/	Challenge bank	Implemented	run_all_challenges.s

Feature	Component	Category	Implementation	Wiring
form-reenables, host/user OOM+suspend, landing-cookie-upload, memory-limits, no-vpn-direct, queue-lifecycle, queue-polling, retry-visible) Challenges submodule (HelixQA challenge framework, p1-f06..f18 CLI-agent feature dirs)		Challenges/	Challenge framework (submodule)	Implemented (submodule) Challenges / catalogue

7. PLANNED “ not yet built (Implementation = Planned)

Confirmed ABSENT from the codebase (grep over docker-compose.yml, dashboard/src, landing/app.py returned no matches “ Â§11.4.6).

Feature	Component	Category	Implementation	Wiring	Tests Coverage	Validation/ Verification	Video- Confirm
Media post-processor service (new container orchestrating transcode/derive pipeline) Web-ready video transcode (browser-playable derivative output) MP3 audio derivation (extract/encode audio track) Post-process status API (report pipeline progress per item) Dashboard pipeline-status UI (surface transcode/	media_postprocessor	Service (planned)	Planned	Not wired “ no compose service exists	None yet	SKIP (no feature to validate)	PENDING “ not y recorded
	media_postprocessor	Pipeline feature (planned)	Planned	Not wired	None yet	SKIP	PENDING “ not y recorded
	media_postprocessor	Pipeline feature (planned)	Planned	Not wired	None yet	SKIP	PENDING “ not y recorded
	media_postprocessor / landing	API feature (planned)	Planned	Not wired “ no route exists in landing/app.py	None yet	SKIP	PENDING “ not y recorded
	dashboard	Web UI feature (planned)	Planned	Not wired “ no component exists	None yet	SKIP	PENDING “ not y recorded

Feature	Component	Category	Implementation	Wiring	Tests Coverage	Validation/ Verification	Video-Confirmation
derive progress) Resume mechanism (resume interrupted/ aborted downloads)	metube / media_postprocessor	Pipeline feature (planned)	Planned	Not wired	None yet	SKIP	PENDING â€” not yet recorded

Reconciliation note (Â§11.4.153 / Â§11.4.118)

Every row above maps to real code (a named compose service, a Flask route in `landing/app.py`, an Angular component/service file under `dashboard/src/app/`, a root CLI script, or a test/challenge file) â€” except the Â§7 PLANNED rows, which are explicitly marked Planned/SKIP because the codebase contains no matching implementation. No code-present feature is omitted; no row lacks corresponding code. Per Â§11.4.6, no verdict claims a PASS without captured evidence, and per Â§11.4.153 no Video-Confirmation cell claims â€œconfirmedâ€” â€” all read PENDING â€” not yet recorded.